



ICE-4101

**Artificial Intelligence and
Neural Networks**

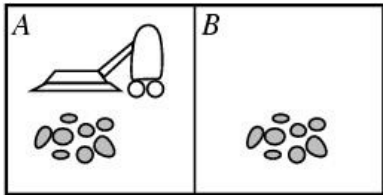
Lecture-6

Types of Agent

Prepared By

Prof. Dr. Md. Ashikur Rahman Khan

The vacuum-cleaner world



```

function REFLEX-VACUUM-AGENT ([location, status]) return an action
  if status == Dirty then return Suck
  else if location == A then return Right
  else if location == B then return Left
  
```

Reduction from 4^T to 4 entries

29 August, 2025

TLo (IRIDIA)

2

Types of Agent



Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:

Five basic kind of agent programs:

- 1) **Simple reflex agents**
- 2) **Model-based reflex agents**
- 3) **Goal-based agents**
- 4) **Utility-based agents**
- 5) **Learning agent**

Prof. Dr. Md. Ashikur Rahman Khan

Simple Reflex/Reflex Agent

The simplest kind of agent is **the simple reflex agent**.

These agents take decisions/**Actions** on the basis of the current percepts and **ignore the rest of the percept history**.

- These agents only succeed in the **fully observable environment**.
- The Simple reflex agent **does not consider any part of percepts history during their decision and action process**.
- **Do not consider the future consequences of their actions**
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action.

Such as a Room Cleaner agent, it works only if there is dirt in the room.

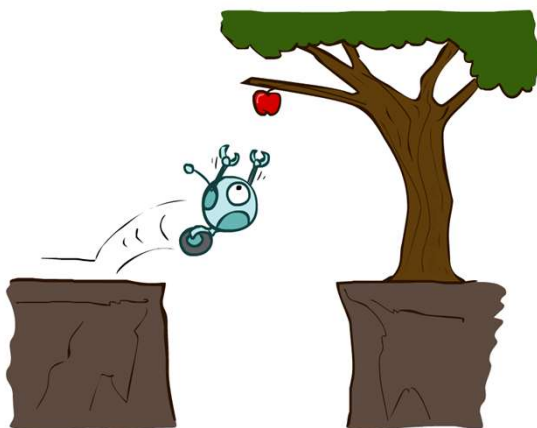
Simple Reflex/Reflex Agent

Reflex agents:

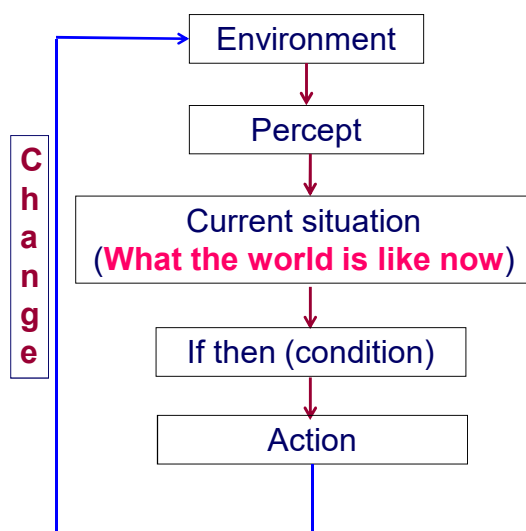
Reflex means
Immediately or
Spontaneously.

**It does not calculate
any calculation just go
to action immediately
is called reflex.**

**Can a reflex agent be
rational?**



Simple Reflex Agent



Simple Reflex Agents

NO MEMORY

**Fails if environment
is partially observable**

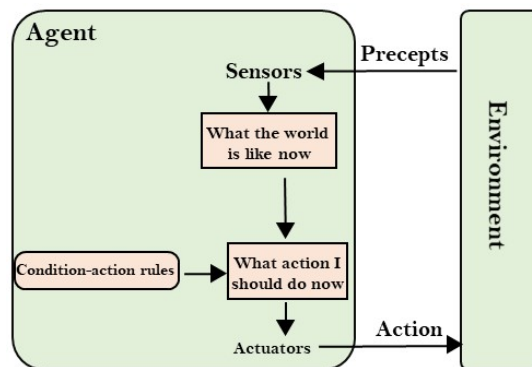


Figure 7: Schematic diagram of a simple reflex agent.

Rectangles represents current internal state
ovals represents the background information

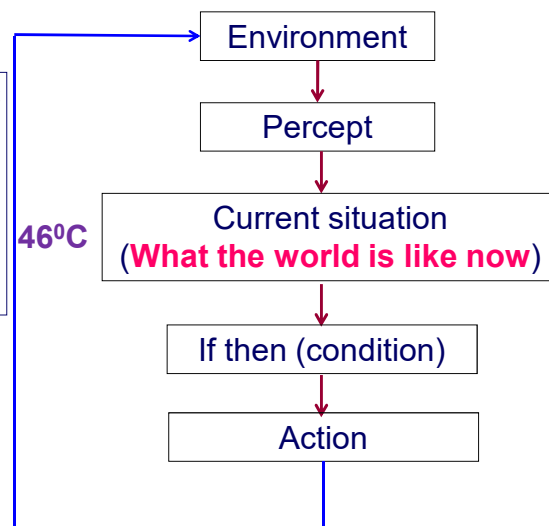
Prof. Dr. Md. Ashikur Rahman Khan

Simple Reflex Agent

Consider **a robot** in a room which performs **to open or close AC**.

Say, when the room temperature is greater than 45°C, then switch on AC.
It's called condition.

**If temp > 45°C
then switch on AC**



Example of Simple Reflex Agent

Vacuum Cleaner:

For example, the vacuum agent whose agent function is tabulated in Figure 2.3 is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt.

An agent program for simple reflex agent (two-state Vacuum Cleaner) is:

Vacuum Cleaner (Continued)

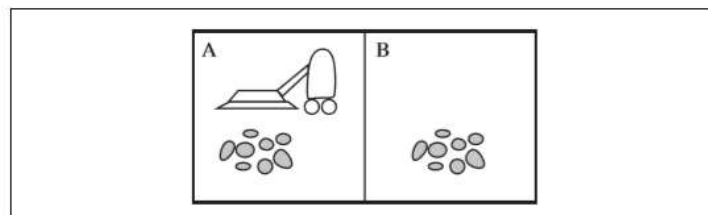


Figure 2.2 A vacuum-cleaner world with just two locations.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Vacuum Cleaner (Continued)

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Example of Simple Reflex Agents

Automated taxi:

Suppose for automated taxi if the condition is “The car in front is braking.” Then, this triggers some established connection in the agent program to the action “initiate braking.”

We call such a connection a **condition–action rule**, written as
if *car-in-front-is-braking* then *initiate-braking*.

The agent program, which is also very simple, is shown in Figure 2.10.

Example of Simple Reflex Agents

It acts according to the rule.

The agent in Figure 2.10 will work *only if the correct decision can be made on the basis of only the current percept—that is, only if the environment is fully observable*.

function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action

static: *rules*, a set of condition-action rules
state, a description of the current world state
action, the most recent action.

state ← INTERPRET-INPUT(*percept*)
rule ← RULE-MATCH(*state*, *rule*)
action ← RULE-ACTION[*rule*]
 return *action*

Will only work if the environment is fully observable
otherwise infinite loops may occur.

13

Advantages and Limitations or Problems Simple Reflex Agents

Advantages:

- Fast and efficient for simple tasks.
- Easy to implement.

Limitations:

- Cannot handle partial observability.
- Cannot learn or plan.
- Ineffective in dynamic or complex environments.

Problems for the simple reflex agent design approach:

- They have very limited intelligence
- They do not have knowledge of non-perceptual parts of the current state
- Mostly too big to generate and to store.
- Not adaptive to changes in the environment.

Model-based Reflex Agents



- ❑ **Model based** means **knowledge based**. Knowledge is gained from history (previous perception).

The human being, we keep in our brain what we have learnt from past this is the **model**. Like Human Brain. We learnt what we have passed all. This the knowledge based.

- ❑ It works on current perception plus past perception.
- ❑ It is also reflex agent, but it consider the past history.
- ❑ This model (knowledge) is used to take decision.
- ❑ On the other hand, simple reflex agent takes decision based what it observes.

Prof. Dr. Md. Ashikur Rahman Khan

Model-based Reflex Agents

The Model-based agent can work in a partially observable environment, and track the situation.

- A model-based agent has two important factors:
 - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
 - **Internal State:** It is a representation of the current state based on percept history.

These agents have the model, "which is knowledge of the world" and based on the model they perform actions.

- Updating the agent state requires information about:
 - How the world evolves
 - How the agent's action affects the world.

Model-based Reflex Agents



Partially Observable Environment (where the agent has no complete knowledge of environment).

Store percept history (Internal Model).

It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

Model the state of the world by:
modeling how the world changes
how its actions change the world

- This can work even with partial information
- It's unclear what to do without a clear goal

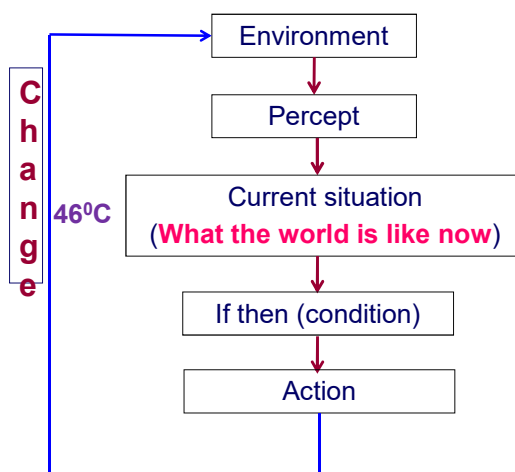
Prof. Dr. Md. Ashikur Rahman Khan

Model-based Reflex Agents

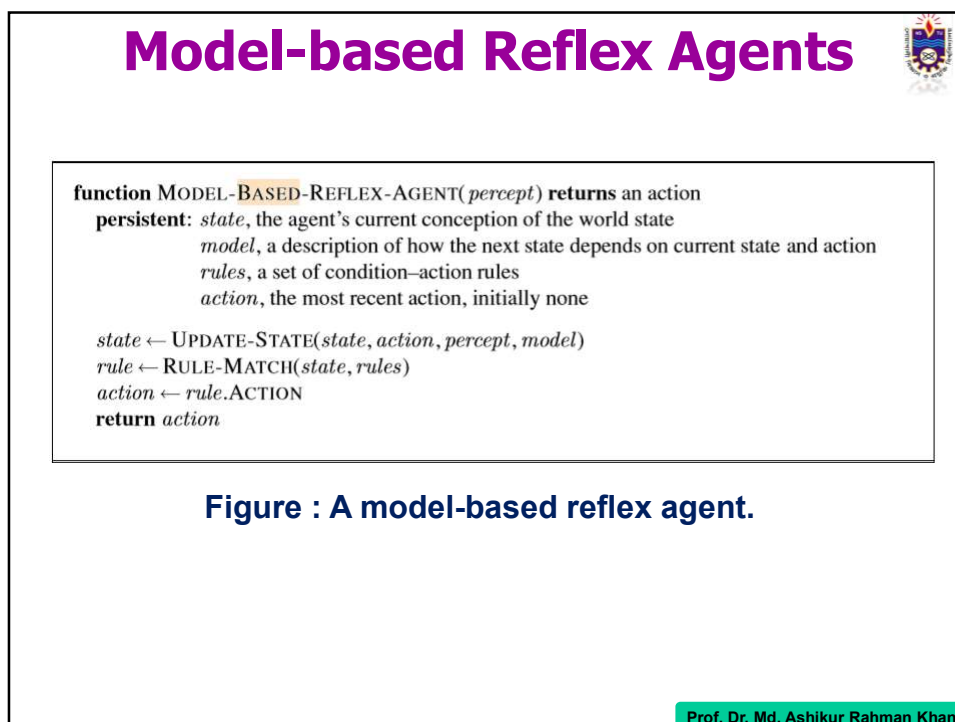
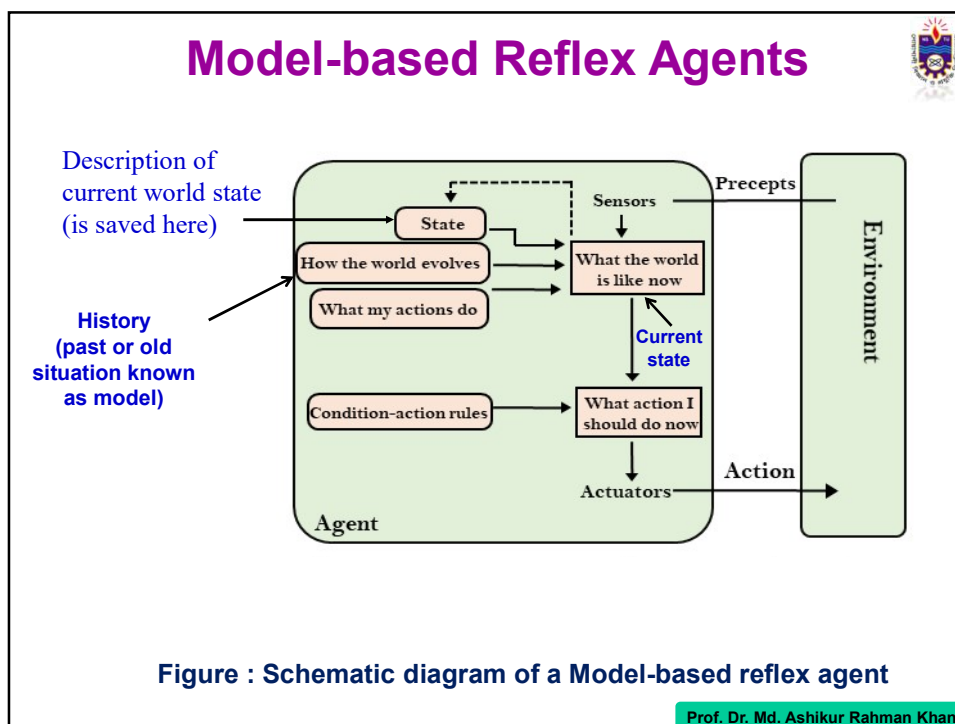
Now if it is that there is no people but the agent will also switch on when the temperature is greater than 45°C. Or

There is some people who doesn't need to switch ON or he feels comfortable but still the agent will switch ON AC.

For this circumstances we need model based agent.



**If temp > 45°C
then switch on AC
If Room is not empty**



Model-based Reflex Agents

Example:

A robotic vacuum remembering which areas have already been cleaned.

Advantages:

- Can handle partially observable environments.
- Retains state and can plan short-term.

Limitations:

- More complex to implement.
- Requires accurate modeling.

Difference between Simple Reflex and Model Based Agent



The simplest agents discussed in Chapter 2 were the reflex agents, which base the iractions on a direct mapping from states to actions. Such agents cannot operate well in environments for which this mapping would be too large to store and would take too long to learn.

Goal-based agents, on the other hand, consider future actions and the desirability of their outcomes.

Goal-based Agents

- Knowing something about the current state of the environment **is not always enough to decide what to do.**
- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- **They choose an action, so that they can achieve the goal.**
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not.

Such considerations of different scenario are called **searching and planning**, which makes an agent proactive.

Goal-based Agents



- Focus on goal. Goal information guides agent's actions (looks to the future)
- **With a current state description, the agent needs some sort of goal information** that describes situations that are desirable. This **Desirable situation is known as goal.**
- The agent program can combine this with information about the results of possible actions in order to choose actions that achieve the goal
- Sometimes achieving goal is simple e.g. from a single action. Other times, **goal requires reasoning about long sequences of actions**
- **This agent advances considering its goal parallel with its current and past percept like human.**
This technique (entire process) is based on searching and planning.
- Usually requires **search** and **planning**
- **To reach the goal first it needs planning.**

Prof. Dr. Md. Ashikur Rahman Khan

Goal-based Agents

Properties of Goal Based Agent:

- Expansion of model based reflex agent
- Desirable Situation (Goal)
- Searching and Planning

Searching and Planning:

Suppose we plan to travel or to go to some places. What we do?

First we search the road, shortest, easiest, secured road.

Then we plan how we will go by bus by motor bike, schedule, stopping point etc.

Model-based Reflex Agents

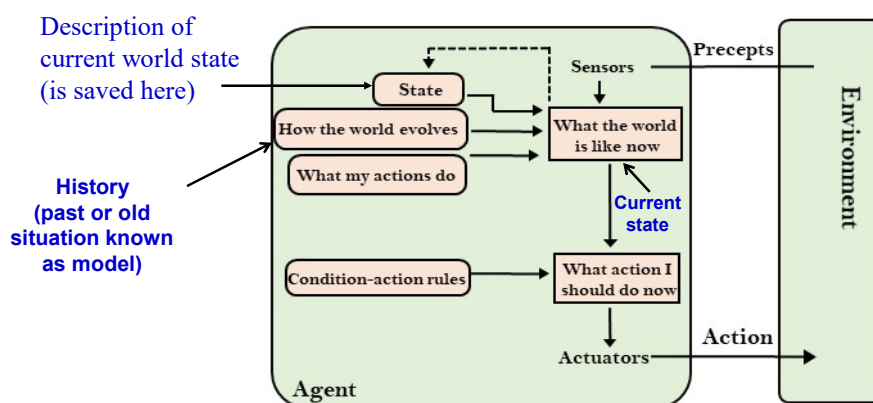
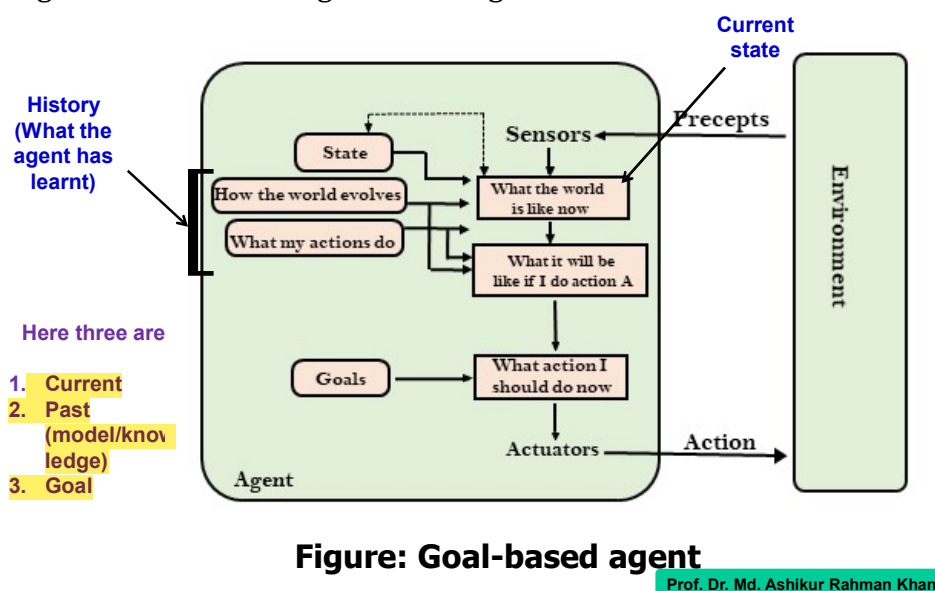


Figure : Schematic diagram of a Model-based reflex agent

Prof. Dr. Md. Ashikur Rahman Khan

Goal-based Agents

Figure 2.13 shows the goal-based agent's structure.



Goal-based Agents

Example:

Amazon — It has made so many intelligent robot.

Chinese company Ali baba has made G+ robot for packet delivery to home (still in testing).

For example, at a road junction, the taxi can turn left, turn right, or go straight on.

The correct decision depends on where the taxi is trying to get to.

Goal-based Agents

Example:

- A path-finding robot that plans the shortest path to a goal.

Advantages:

- More flexible and adaptable.
- Can make decisions based on outcomes.

Limitations:

- Computationally intensive.
- Requires goal formulation and planning algorithms.

Goal-based Agents



The **reflex agent brakes** when it sees brake lights.

On the other hand, **a goal-based agent**, could reason that if the car in front has its brake lights on, **it will slow down**.

Given the way the world usually evolves, the only action that will **achieve the goal of not hitting other cars** is to brake.

Although the goal-based agent appears less efficient, it is more flexible because the knowledge that supports its decisions is represented explicitly and can be modified.

If it starts to rain, the agent can update its knowledge of **how effectively its brakes will operate**; this will automatically cause all of the relevant behaviors to be altered to suit the new conditions.

Prof. Dr. Md. Ashikur Rahman Khan

Goal-based Agents



For the reflex agent, on the other hand, we would have to rewrite many condition–action rules.

The goal-based agent's behavior can easily be changed to go to a different destination, simply by specifying that destination as the goal.

The reflex agent's rules for when to turn and when to go straight will work only for a single destination; they must all be replaced to go somewhere new.

Prof. Dr. Md. Ashikur Rahman Khan

Goal-based Agents



Search (Solving Problems by Searching—Chapter 3, Beyond Classical Search—Chapter 4, Adversarial Search—Chapter 5) and

Planning (Classical Planning—Chapters 10; and Planning and Acting in the real world—Chapter 11)

are the subfields of AI devoted to finding action sequences that achieve the agent's goals.

Prof. Dr. Md. Ashikur Rahman Khan



Vacuum-cleaner world

Goal-based agents vs reflex-based agents

- Although goal-based agents appears less efficient, it is more flexible because the knowledge that supports its decision is represented explicitly and can be modified
- On the other hand, for the reflex-agent, we would have to rewrite many condition-action rules
- The goal based agent's behavior can easily be changed
- The reflex agent's rules must be changed for a new situation

Prof. Dr. Md. Ashikur Rahman Khan



Utility-Based Agent

Goals alone are not enough to generate high-quality behavior in most environments.

For example, many action sequences will get the taxi to its destination (thereby achieving the goal) **but some are quicker, safer, more reliable, or cheaper than others.**

Goals just provide a crude binary distinction between “happy” and “unhappy” states.

A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent.

Because “happy” does not sound very scientific, economists and computer scientists use the **term utility** instead.

Prof. Dr. Md. Ashikur Rahman Khan

Utility-Based Agent



It is the extension of goal-based agent.

Some solutions to goal states are better than others.

Which one is best is given by a utility function.

Which combination of goals is preferred?

These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.

- UBA act **based not only goals but also the best way to achieve the goal.**
- UBA is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

Prof. Dr. Md. Ashikur Rahman Khan

Utility-Based Agent

Real life Example:

Consider GPS System Tracking system.

Suppose you will travel state to goal around 350 km.

GPS will show two or three roads however, by default it will show the shortest way/road. **You becomes in happy state.**

When you follow this shortest way and driving after certain distance you see that road at blocked condition/situation due to any accident. At this state what will be happened, you will not arrive to the goal state at time and it will take more time. Consequently, **you reach at unhappy state.**

Now the GPS will try to find out any alternate way or further more shortcut way to reach the goal.

Utility-Based Agent



Furthermore, in two kinds of cases, goals are inadequate but a utility-based agent can still make rational decisions.

- ❑ **First**, when there are conflicting goals, only some of which can be achieved (for example, speed and safety), the utility function specifies the appropriate tradeoff.
- ❑ **Second**, when there are several goals that the agent can aim for, none of which can be achieved with certainty, utility provides a way in which the likelihood of success can be weighed against the importance of the goals.

Prof. Dr. Md. Ashikur Rahman Khan

Utility-Based Agent

Improve on goals:

- Selecting between conflicting goals
- Select appropriately between several goals based on likelihood of success.

The properties of Utility Based Agent

- **Focus on Utility not on Goal**
- **Utility Function**
- **Deals with Happy and Unhappy state**

i.e. after performing the action the agent reaches the state, is this state happy state or unhappy state!

This happiness or unhappiness of state is measured by utility function.

It works on partially observable environment.

Utility-based Agents

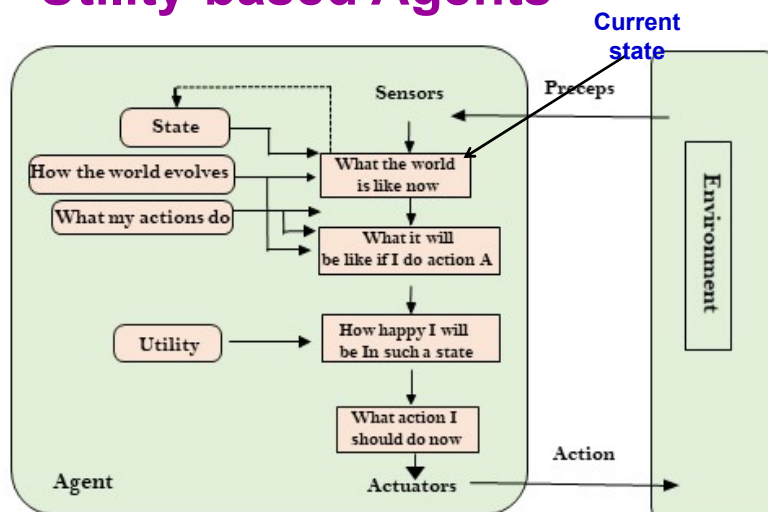


Figure : Schematic diagram of a Utility-based agent

Utility-Based Agent

- Utility means how much beneficial
- What if there are many paths to the goal?
- Utility measures which states are preferable to other state
- Maps state to real number (utility or "happiness")
- Goals alone are not really enough to generate high quality behavior in most environments – they just provide a binary distinction between happy and unhappy states
- A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent if they could be achieved
- Happy – Utility (the quality of being useful)
- A utility function maps a state onto a real number which describes the associated degree of happiness

Prof. Dr. Md. Ashikur Rahman Khan

Utility-Based Agent

Example:

- An AI system that selects the safest and fastest route with minimal fuel usage.

Advantages:

- Can handle trade-offs and uncertainty.
- Provides a richer decision-making model.

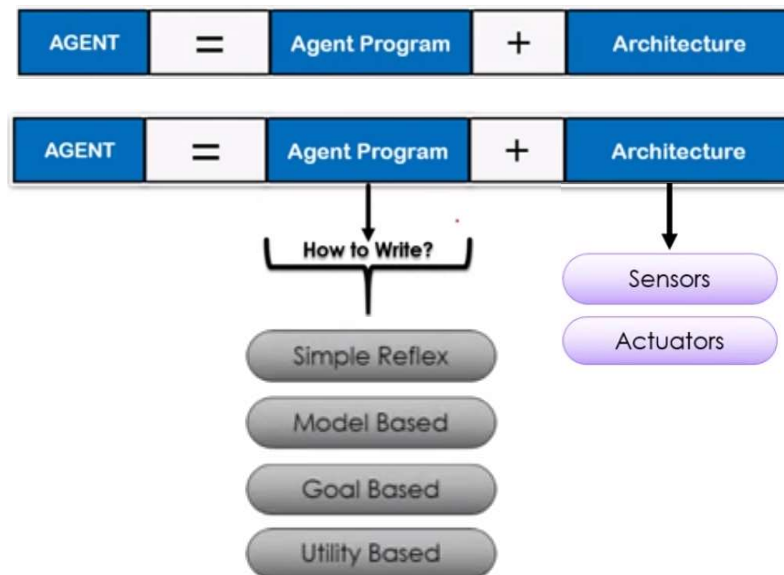
Limitations:

- Defining a good utility function can be difficult.
- Requires probabilistic reasoning and modeling.

Utility-Based Agent

Partial observability and stochasticity are ubiquitous in the real world, and so, therefore, is decision making under uncertainty. Technically speaking, a rational utility-based agent chooses the action that maximizes the expected utility of the action outcomes—that is, the utility the agent expects to derive, on average, given the probabilities and utilities of each

Learning Agent



Learning Agent

A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.

- It starts to act with basic knowledge and then able to act and adapt automatically through learning.

- Learning has another advantage, it allows the agent to operate in initially unknown environments and to become more competent than its initial knowledge alone might allow.

- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.

Learning Agent

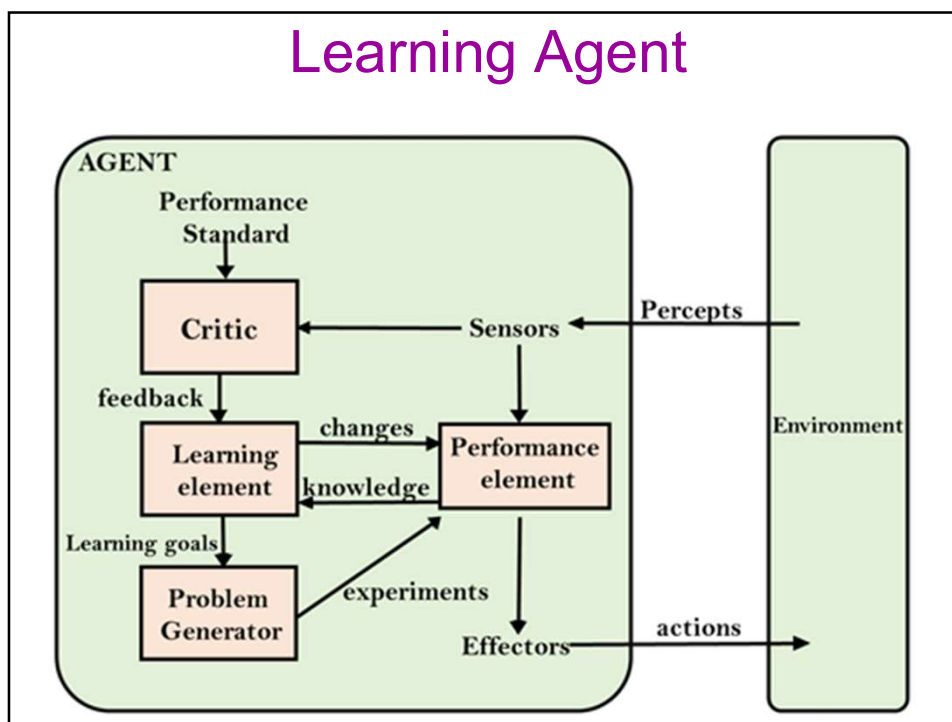
A learning agent can be divided into four conceptual components,

- Learning Element
- Performance Element
- Critic
- Problem Generator

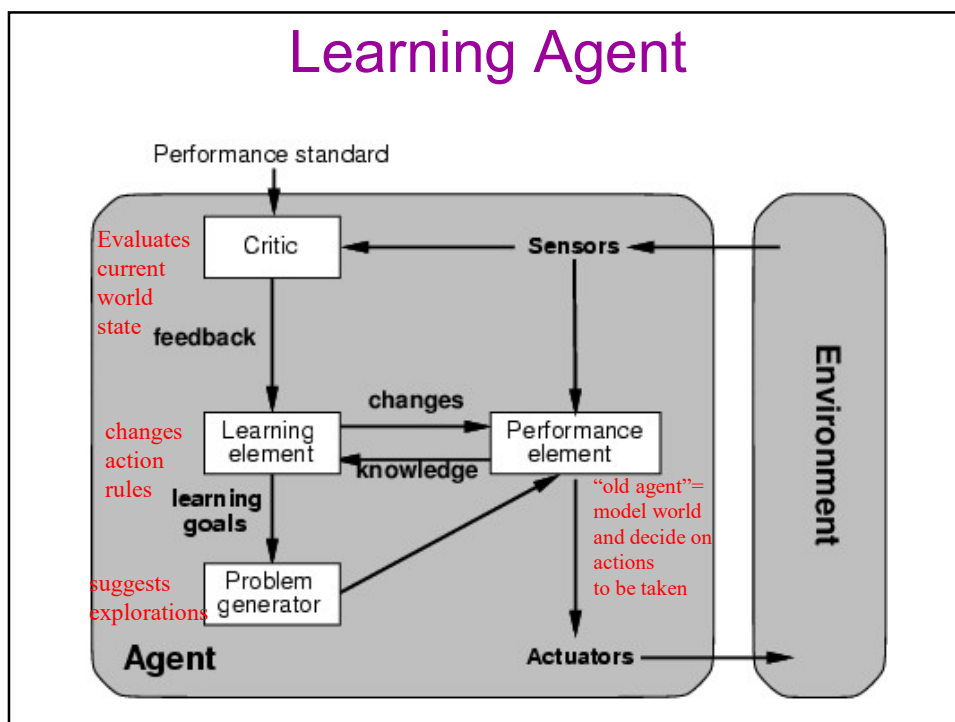
A learning agent has mainly four conceptual components, which are:

- **Learning element:** It is responsible for making improvements by learning from environment
- **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
- **Performance element:** It is responsible for selecting external action
- **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Learning Agent



Learning Agent



Learning Agent

How does an agent improve over time?

By monitoring it's performance and suggesting better modeling, new action rules, etc.

- All previous agent-programs describe methods for selecting *actions*.
 - Yet it does not explain the origin of these programs.
 - Learning mechanisms can be used to perform this task.
 - Teach them instead of instructing them.
 - Advantage is the robustness of the program toward initially unknown environments.

Learning Agent

The design of the learning element depends very much on the design of the performance element.

When trying to design an agent that learns a certain capability, the first question is not “How am I going to get it to learn this?” but “What kind of performance element will my agent need to do this once it has learned how?”

Given an agent design, learning mechanisms can be constructed to improve every part of the agent.

The critic tells the learning element how well the agent is doing with respect to a fixed performance standard.

The critic is necessary because the percepts themselves provide no indication of the agent's success.

Learning Agent

For example, a chess program could receive a percept indicating that it has checkmated its opponent, but it needs a performance standard to know that this is a good thing; the percept itself does not say so.

4. **Problem Generator**: The last component of the learning agent is the problem generator. It is responsible for suggesting actions that will lead to new and informative experiences.

Exploration vs. exploitation

Let us return to the automated taxi example. The performance element consists of whatever collection of knowledge and procedures the taxi has for selecting its driving actions. The taxi goes out on the road and drives, using this performance element. The critic observes the world and passes information along to the learning element.

Learning Agent

For example, after the taxi makes a quick left turn across three lanes of traffic, the critic observes the shocking language used by other drivers. From this experience, the learning element is able to formulate a rule saying this was a bad action, and the performance element is modified by installation of the new rule. The problem generator might identify certain areas of behavior in need of improvement and suggest experiments, such as trying out the brakes on different road surfaces under different conditions.

Learning Agent

For example, if the taxi exerts a certain braking pressure when driving on a wet road, then it will soon find out how much deceleration is actually achieved. Clearly, these two learning tasks are more difficult if the environment is only partially observable.

The forms of learning in the preceding paragraph do not need to access the external performance standard—in a sense, the standard is the universal one of making predictions that agree with experiment. The situation is slightly more complex for a utility-based agent that wishes to learn utility information.

For example, suppose the taxi-driving agent receives no tips from passengers who have been thoroughly shaken up during the trip. The external performance standard must inform the agent that the loss of tips is a negative contribution to its overall performance; then the agent might be able to learn that violent maneuvers do not contribute to its own utility. In a sense, the performance standard distinguishes part of the incoming percept as a reward (or penalty) that provides direct feedback on the quality of the agent's behavior.

Learning Agent

learning methods. There is, however, a single unifying theme. Learning in intelligent agents can be summarized as a process of modification of each component of the agent to bring the components into closer agreement with the available feedback information, thereby improving the overall performance of the agent.

Learning Agent

Example:

- A recommendation system improving suggestions based on user feedback.

Advantages:

- Highly adaptive and capable of improving in unknown environments.
- Suitable for dynamic and evolving environments.

Limitations:

- May need large data and time to learn.
- Risk of learning incorrect behavior.

Comparison Table: Agent Types

Agent Type	Memory	Planning	Learning	Environment Suitability
Simple Reflex	✗	✗	✗	Fully observable, static
Model-Based Reflex	✓	✗	✗	Partially observable
Goal-Based	✓	✓	✗	Sequential, dynamic
Utility-Based	✓	✓	✗	Stochastic, uncertain
Learning Agent	✓	✓	✓	Unknown, evolving

Example Use Cases:

- **Bank fraud detection system** → Utility-Based / Learning Agent
- **Traffic light system** → Model-Based Reflex Agent
- **Autonomous warehouse robot** → Goal-Based + Learning Agent
- **Voice-controlled assistant** → Learning + Utility-Based Agent

Example of Simple Reflex Agents

Automated taxi:

Question

- Suppose for automated taxi if the condition is "The car in front is braking." Then, this triggers some established connection in the agent program to the action "initiate braking." Write the Pseudocode for Simple Reflex agent and explain this with automated taxi for this scenario or condition.

We call such a connection a condition–action rule, written as if *car-in-front-is-braking* then *initiate-braking*.

ANSWER

function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action

static: *rules*, a set of condition-action rules

state, a description of the current world state

action, the most recent action.

state $\leftarrow$ INTERPRET-INPUT(*percept*)

rule $\leftarrow$ RULE-MATCH(*state*, *rule*)

action $\leftarrow$ RULE-ACTION[*rule*]

return *action*

57

Question



1. Consider a vacuum-cleaner world shown in Figure 2.2. This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square. A partial tabulation of this agent function is shown in following Figure.

- Find the possible states for this Vacuum Cleaner world problem.
- Write and discuss the Pseudocode for this Vacuum Cleaner when it is a reflex agent or simple reflex agent.

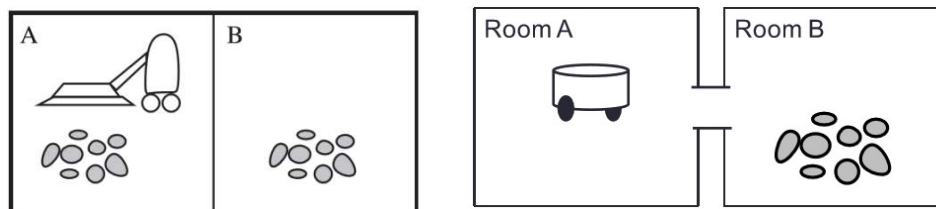


Figure: A vacuum-cleaner world with just two locations.

Prof. Dr. Md. Ashikur Rahman Khan

Future Directions in Agent Research

As AI evolves, research in agent design focuses on:

- **Collaborative multi-agent systems** (e.g., swarm robotics)
- **Emotionally aware agents** (affective computing)
- **Ethical reasoning agents** (AI ethics)
- **General-purpose agents** (AGI)



Prof. Dr. M.A.R. Khan, Dept. of ICE